

The Impact of Nearly Universal Insurance Coverage on Health Care Utilization: Evidence from Medicare

Card, Dobkin, Maestas (2008)

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Roadmap

- 1 Paper Overview & Contribution
- 2 Story & Key Findings
- 3 Data
- 4 Empirical Strategy
- 5 Results
- 6 Conclusion

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Overview of Paper

- $\frac{1}{5}$ of non-elderly uninsured in 2005.
- Due to Medicare eligibility at 65, $< 1\%$ of elderly are uninsured.
- **Research question:** How does gaining near-universal insurance at 65 affect health care utilization?
- **Identification:** RD exploiting sharp age cutoff.
- **Data:** Hospital discharge records from CA, FL, NY and NHIS survey data.

Contribution (Short Discussion)

- **What's new:** Broader range of outcomes than previous work, quantify differences after receiving Medicare between groups
- **Why it matters:** Inform policy debates about expanding insurance coverage.

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Paper's Story / Key Findings

Key qualitative takeaways

- Increase in overall coverage, with lessening coverage gaps.
- Increase in multiple coverage, especially for highly educated, non-minorities.
- Increase in medical care utilization, though effects differ by group

Key quantitative takeaways

- Medicare coverage increases 60 pp at age 65.
- Probability of any insurance coverage increases by 9.5 pp
- Probability of multiple coverage increases by 44.1 pp

Main Results — One-Slide Overview

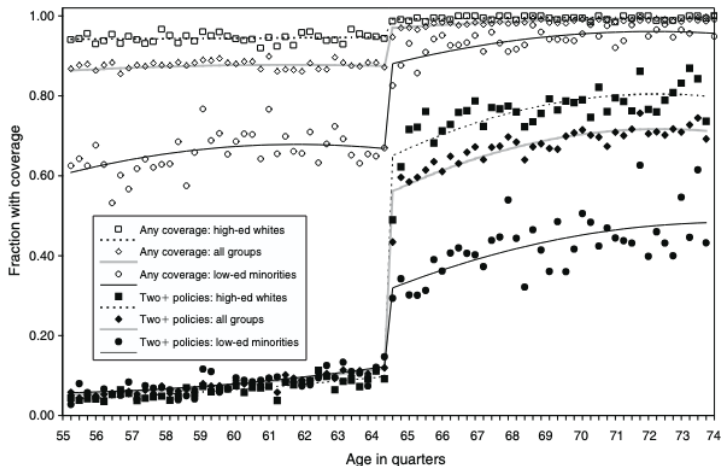


FIGURE 1. COVERAGE BY ANY INSURANCE AND BY TWO OR MORE POLICIES, BY AGE AND DEMOGRAPHIC GROUP

- Sharp increase in Medicare coverage at 65.
- Increase in any insurance coverage at 65.
- Large increase in multiple forms of coverage at 65.

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Data: Sources & Coverage

- National Health Interview Survey (NHIS).
- Hospital discharge records from CA, FL, NY.
- 1992-2003, ages 55-75, 160,821 agents in HIS, 2.8 - 4.0 million in discharge records.

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Reduced-Form Specification

$$y_{ija} = X_{ija}\alpha + f_j(a; \beta) + \sum_k C_{ija}^k \delta^k + u_{ija} \quad (1)$$

- y_{ija} : outcome for individual i in group j at age a .
- X_{ija} : individual, regional, etc. controls
- $f_j(a; \beta)$: smoothing function based on age profile
- C_{ija}^k : characteristics of insurance coverage, can include coverage indicator or other insurance covariates

Coverage Functions

$$C_{ija}^1 = X_{ija}\beta_j^1 + g_j^1(a) + D_a\pi_j^1 + v_{ija}^1 \quad (2)$$

$$C_{ija}^2 = X_{ija}\beta_j^2 + g_j^2(a) + D_a\pi_j^2 + v_{ija}^2 \quad (3)$$

- β_j^1, β_j^2 : group-specific coefficients.
- $g_j^1(a), g_j^2(a)$: smoother for group-specific age profiles.
- D_a : indicator for age 65+.

Final Reduced Form

$$y_{ija} = X_{ija}(\alpha_j + \beta_j^1 \delta_j^1 + \beta_j^2 \delta_j^2) + h_j(a) + D_a \pi_j^y + \nu_{ija}^y \quad (4)$$

- $h_j(a)$: reduced form age profile.
- π_j^y : aggregated effect of Medicare eligibility on outcome y .
- ν_{ija}^y : new error term.

Assuming profiles $f_j(a), g_j^1(a), g_j^2(a)$ are all continuous at 65, any discontinuity in y at 65 is due to Medicare eligibility.

Overcoming Pitfalls

- **Lack of Data** Unable to clearly observe information on insurance packages of each individual
- **Solution:** Use at least two measures of coverage to proxy for insurance generosity.
- **Endogeneity of Coverage** Insurance coverage is all that matters, meaning the effect will differ by group.
- **Solution:** Estimate separate age profiles and coverage functions by group.

$$\pi_j^y = \delta^0 + \delta^1 \pi_j^1 + \delta^2 \pi_j^2 + e_j \quad (5)$$

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Table 1 — Insurance Characteristics Just Before Age 65

	On Medicare		Any insurance		Private coverage		2+ Forms		Managed care	
	63–64	RD@65	63–64	RD@65	63–64	RD@65	63–64	RD@65	63–64	RD@65
Overall sample	12.3	59.7	87.9	9.5	71.8	–2.9	40.8	44.1	59.4	–28.4
<i>Classified by ethnicity and education:</i>										
White non-Hispanic, HS dropout	21.1	58.5	84.1	13.0	63.5	–6.2	15.0	34.3	48.1	–25.0
White non-Hispanic, HS graduate	11.4	64.7	92.0	7.6	80.5	–1.9	19.7	31.1	51.8	–15.4
White non-Hispanic, $\geq$ some college	6.1	68.4	94.6	4.4	85.6	–2.3	8.8	35.1	69.1	–40.1
Minority, HS dropout	19.5	44.5	66.8	21.5	33.2	–1.2	11.4	19.4	39.1	–8.3
Minority, HS graduate	16.7	44.6	85.2	8.9	60.9	–5.8	13.6	23.4	54.2	–15.4
Minority, $\geq$ some college	10.3	52.1	89.1	5.8	73.3	–5.4	11.1	38.4	66.2	–22.3
<i>Classified by ethnicity only:</i>										
White non-Hispanic (all)	10.8	65.2	91.8	7.3	79.7	–2.8	10.4	51.9	61.9	–33.6
Black non-Hispanic (all)	17.9	48.5	84.6	11.9	57.1	–4.2	13.4	27.8	48.2	–13.5
Hispanic (all)	16.0	44.4	70.0	7.3	42.5	–2.0	10.8	21.7	52.9	–12.1

Note: Percentages at ages 63–64 shown in odd-numbered columns; even-numbered columns are RD estimates at 65 from linear probability models with quadratic age controls and covariates for gender, race/ethnicity, education, region, and sample year. Standard errors in parentheses in the original table.

Table 2 — Employment Discontinuities at Age 65

	Data from NHIS		Data from March CPS			
	Employed	Full time	Employed	Hours/Wk	Retired	Earnings (1000s)
Overall sample	0.3	0.8	1.8	1.3	-1.6	-0.2
White, HS dropout	-0.6	-0.5	-1.2	-0.7	2.0	-0.4
White, HS grad	0.4	0.9	2.0	1.6	-1.7	-0.3
White, $\geq$ some college	0.8	1.3	2.6	1.8	-2.3	-0.5
Minority, HS dropout	-0.2	-0.6	-0.7	-0.5	1.0	-0.1
Minority, HS grad	0.1	0.4	1.5	0.9	-1.1	-0.2
Minority, $\geq$ some college	0.6	0.8	2.2	1.4	-1.9	-0.3

Note: RD estimates at age 65 with quadratic age controls. Controls include gender, race/ethnicity, education, region, and sample year. SEs clustered by quarter (cols 1–2) or year (cols 3–6).

Figure 2 - Employment Rates by Age and Demographic Group

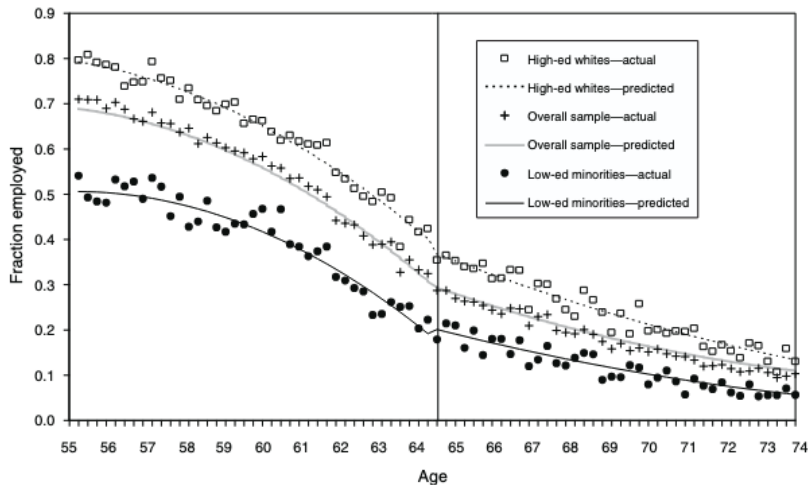


FIGURE 2. EMPLOYMENT RATES BY AGE AND DEMOGRAPHIC GROUP (1992–2003 NHIS)

Table 3 — Access to Care and RD at 65

	1997–2003 NHIS				1992–2003 NHIS			
	Delay (63–64)	RD@65	No care (63–64)	RD@65	Saw doctor (63–64)	RD@65	Hosp. stay (63–64)	RD@65
Overall sample	7.2	-1.8	4.9	-1.3	84.8	1.3	11.8	1.2
White, HS dropout	9.1	-2.5	6.7	-1.7	81.5	1.5	10.9	1.4
White, HS grad	6.8	-1.6	4.3	-1.1	86.7	1.1	12.2	1.0
White, $\geq$ some college	5.4	-1.2	3.8	-0.9	88.9	1.0	11.4	0.8
Minority, HS dropout	12.5	-3.0	9.2	-2.0	75.3	1.9	10.7	1.7
Minority, HS grad	8.3	-2.1	5.6	-1.5	83.0	1.4	11.6	1.3
Minority, $\geq$ some college	6.9	-1.7	4.7	-1.2	86.2	1.2	12.0	1.1

Note: Odd columns show means for ages 63–64; even columns are RD estimates at 65. Models include linear/quadratic covariates. SEs in parentheses in source table.

Table 4 — Hospital Admissions and Insurance at 65

	All		Whites		Hispanics		Blacks	
	Rate 60–64	RD@65	Rate 60–64	RD@65	Rate 60–64	RD@65	Rate 60–64	RD@65
All admissions	1443	7.57	1407	7.74	1262	9.47	2008	4.39
ER admission	761	3.30	688	3.70	774	2.63	1313	1.93
Non-ER admission	682	12.16	718	11.51	488	19.89	695	8.92
Heart disease	220	8.40	215	8.90	203	9.60	310	5.80
Hip fracture	95	5.90	90	6.20	72	7.50	118	4.10
Probability of coverage	82.7	15.0	86.7	12.7	69.1	20.3	79.0	17.6

Note: Top rows are fractions insured; bottom rows are RD estimates in log admissions at 65. Quadratic age controls included. SEs in parentheses in source table.

Figure 3 - Hospital Admission Rates by Age and Ethnicity

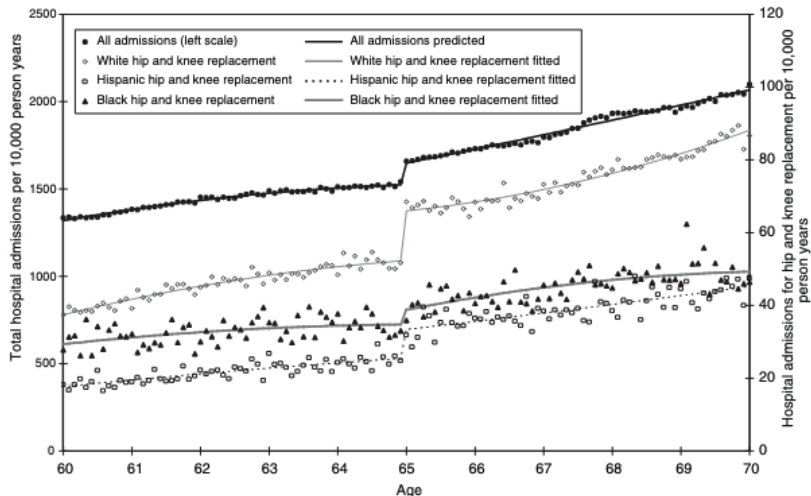


Table 5 — Summary of Effects on Disparities

	Coeff. on cov. RD	R^2	Low-ed min.–Hi-ed W	Black–White	Hisp.–White	% change: Low-ed min.–Hi-ed W	% change: Black–White	% change: Hisp.–White
<i>Panel A: Based on change at 65</i>								
Delay in care last year	-0.19	0.72	7.6	5.2	6.8	-42.8	-36.0	-29.4
No care last year	-0.12	0.39	8.0	4.9	6.0	-25.6	-24.5	-19.0
Saw doctor last year	0.07	0.31	-10.5	-8.2	-7.6	18.0	15.5	13.9
Hospital stay last year	0.11	0.44	2.4	1.9	1.7	45.8	36.5	32.0
<i>Panel B: Based on multiple coverage at 65</i>								
Total admissions	0.03	0.74	230	724	-193	3.9	-0.1	5.8
Hip/knee replacement	0.05	0.66	12	41	-9	4.1	3.6	3.3
Heart failure	0.07	0.52	21	65	-14	9.0	7.4	6.5

Note: Col (1) coefficients are from regressions of RD outcomes on coverage RD. Col (2) shows R^2 . Cols (3–5) show coverage. SEs in parentheses in source.

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Conclusion

What we learned:

- Medicare eligibility at 65 leads to large increases in healthcare utilization, especially for minorities
- Medicare leads to increased insurance coverage, especially multiple coverage
- admissions for discretionary procedures increase more than emergency admissions across all groups, though is particularly pronounced for Hispanics

Thank You

Questions?